

10th International Conference of Information and Communication Technology (ICICT-2020)

Virtual instrument based on electronic and electrician's experiment teaching laboratory design

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Abstract

The improvement of teaching quality in order to promote the electronic electrician experiment, training students' ability of engineering quality and the position, the paper analyzes the current situation of electrical and electronics experimental teaching and the existing problems, this paper expounds the necessity of constructing the virtual instrument laboratory, and from two aspects of hardware modules and software system design construction scheme for the overall design.

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Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: electrical and electronics; experiment teaching; virtual instrument; design

1. Introduction

Local colleges and universities currently used in electrical and electronic teaching instruments mainly is based on integrated circuit digital instrumentation, such as multimeter, signal generator, oscilloscope, etc. These instruments test functional, high testing precision, can satisfy the basic experiment teaching of the basic requirements. But, with the continuous development of electronic information technology, on the one hand, with the microprocessor as the core of intelligent instrument and meter constantly, and technology updates faster; On the other hand, the automatic test technology has been widely applied in the field of instrument testing, a large number of automatic test equipment, change the previous way of equipment maintenance, at the same time for the science and technology quality of the equipment maintenance personnel put forward higher request^[1-2]. Therefore, if continue to use the traditional experimental teaching

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mode, is bound to disconnect with the current engineering application technology, unfavorable to the cultivation of students' science and technology quality and the improvement of professional and technical personnel working ability. Virtual instrument is put forward to breakthrough the traditional concept of instrument, represents the trends of instrumentation development, it combines computer technology and network technology and testing technology of remote measuring and remote control become an easy thing. Therefore, the application of virtual instrument in electrical and electronic teaching can not only lower cost to let the students come into contact with the latest equipment, and can cultivate the students to keep pace with The Times of science and technology literacy and the ability to work demand oriented.

2. Functional Analysis

The current electrical and electronic laboratory are mostly with various curriculum independent experiment box and some general traditional instruments built experimental platform, difficult to achieve function extension and resource sharing, usually only a few demonstration and validation experiments, unable to reflect the development and changes of the current technology, can not effectively stimulate students' interest in learning and cultivate the students' innovation ability. Introduction of virtual instrument technology in electrical and electronic experimental teaching, can make up for the inadequacy of traditional experimental teaching mode, stimulate students' learning enthusiasm and innovative consciousness, and the next course practice and design process has a strong role in promoting and pushing. Virtual instrument laboratory should have the following functions:

(1) Laboratory staff can according to the needs of the experiment design of virtual instrumentation. Such as signal generator, spectrum analyzer, frequency sweep meter, oscilloscope, etc^[3];

(2) The operator can according to the experiment requirement, through the friendly human-computer interaction environment himself interested in the experiment;

(3) To produce on the experiment data and image recording, analyzing, fill out the electronic experimental report, and has a perfect lab report management functions;

(4) Make full use of the Internet, the campus network, the advantage of the characteristics of virtual instrument digital, realize the resource sharing, laboratory instruments make long-distance experiment, laboratory more collaborative new concepts such as large experiment is possible;

(5) With experimental teaching other auxiliary functions, such as experiment guidance function (experiment purpose, the use of equipment information, the principle diagram of the experiment, experiment contents, summary), laboratory equipment, instrument, teachers online FAQ online help function, etc.

Through the function analysis, the virtual instrument laboratory as electrical and electronic experimental teaching platform, is also a virtual instrument design platform, experimental resources sharing platform, experimental equipment use and interactive control platform.

3. Hardware Modules

3.1. Modules Composition

Virtual instrument laboratory for electrical and electronic experimental teaching, especially the instrument using curriculum, make full use of the virtual instrument technology, network technology, bus technology, testing technology, modular data acquisition circuit, etc., the module composition block diagram is shown in Figure 1.

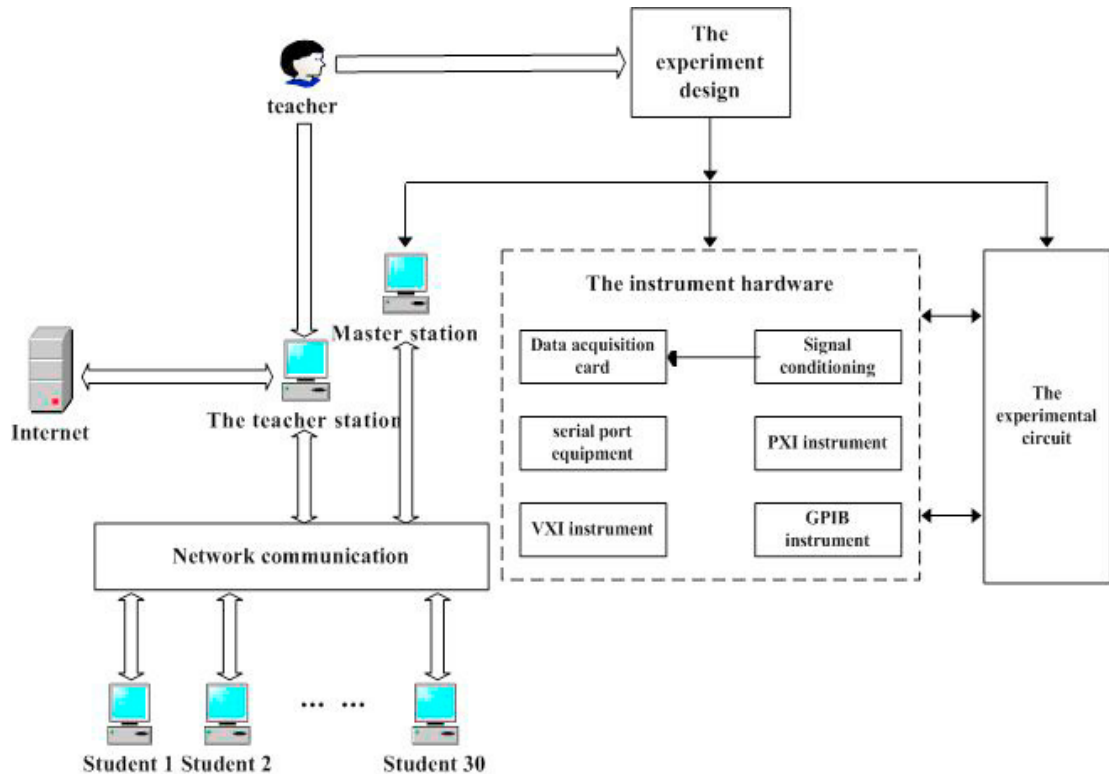


Fig. 1. Hardware construction of virtual instrument laboratory.

3.2. The Instrument Hardware

3.2.1. The Signal Conditioning Module

The signal conditioning circuit is the main purpose of the acquisition card and tested the experimental circuit and realization of electrical isolation between the acquisition channel, and the D/A output signal power to amplify.

3.2.2. The Data Acquisition Card

The data acquisition card is a kind of built-in function based on computer standard bus card, plug directly into the computer PCI slot, A/D, D/A conversion, digital I/O and counter/timer function.

3.2.3. Serial Port Equipment

Serial port equipment mainly includes traditional RS232 serial bus, USB universal serial bus (USB) instrument instrument. Due to the serial port can only directly communicate with a peripheral, so a serial port equipment limited in application, is suitable for the requirements of virtual instrument in low or experiment test system.

3.2.4. PXI Instrument

The PXI expanded PCI in the field of instrument, the chassis (power), PXI controller and PXI/Compact PCI modules. PXI have a high scalability, with eight expansion slots, and the product is fully compatible with the PCI bus. Due to the expansion of the PCI bus for PXI instruments in the field of advantage with computer excellent cost-effective, are likely to become the future mainstream of experimental teaching platform for virtual instrument.

3.2.5. VXI Instrument

VXI system hardware is mainly composed of main box, "O tank" of controller and module instrument that has a variety of functions. Function modules of the system can be arbitrary replacement, plug and play of the new system, because of VXI module instrument prices are relatively high, at present is mainly used for speed, high accuracy requirement of large and medium scale test system.

3.2.6. GPIB Instrument

GPIB system is to insert a GPIB interface card in the microcomputer, through 24 or 25 line cable connection to the instrument GPIB interface, a kind of GPIB interface to connect 14 instrument, many sets of instruments can be easily combined, component test system. Because the GPIB system function and customizable instrument panel, upgrade, easy development, use, maintenance is convenient, at present there are still more application in experimental teaching.

3.3. Master Station

Console is the core unit of the laboratory, experimental design, the virtual instrument design and data distribution, etc. Experimental design mainly includes the experiment purpose, requirement, design of experiment content; Virtual instrument design is refers to the development of virtual instrument in accordance with the contents of design of experiment module; Major instrument hardware data distribution function and the data such as the network share of resources.

3.4. Teachers Station

Teachers with the realization of teachers' role in the local area network, realize the interaction with students, through the network to complete the course of experiment teaching, experiment of the track coach and corrects the experiment report, and the experimental results of the upload.

3.5. Students Station

Students with students to complete the experiment platform of virtual instrument, the same of students and teachers are realized by web to finish the experiment course of study, the solution of the problems and the experiment report submitted, through interactions with master station a virtual instrument module load and access to experimental data and other resources.

4. The Software System Composition

4.1. The Software System Framework

Experiment teaching platform based on virtual instrument technology in hardware was determined, can through the software to implement different functions, all the same access to the master station, through the network implementation measurement, analysis and processing of experimental data. On the basis of laboratory hardware system, combined with the function of each composition module, generalize a laboratory analysis software system structure, as shown in figure 2.

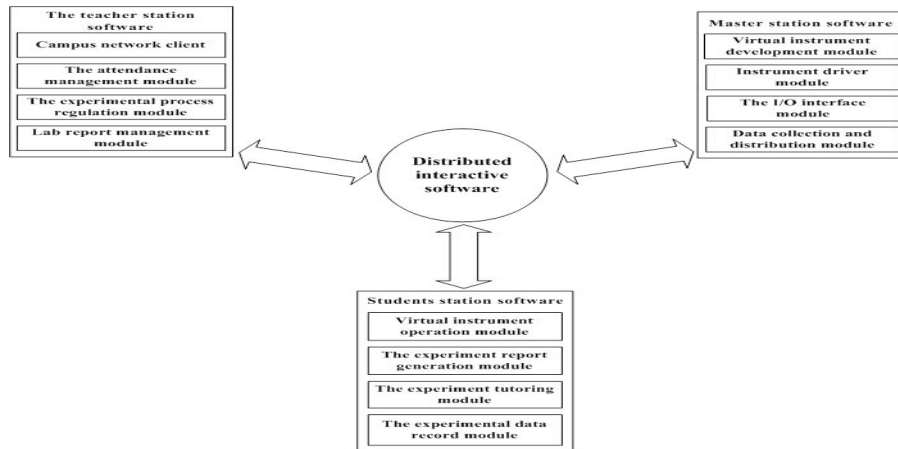


Fig. 2. Software construction of virtual instrument laboratory

4.2. Master Station Software

Master station software is decided to virtual instrument laboratory functions and running effect of the core part, is the key to the realization of hardware resource sharing and data acquisition, mainly includes the following several parts^[4];

- (1) Virtual instrument development module: mainly provides an integrated development environment, with a high-level interface to the hardware of the instrument;
- (2) Instrument driver module, is to achieve one of the most important part of the test, is used to implement instrument hardware communication and control functions;
- (3) The I/O interface module: as the basis of virtual instrument system software, computer and instrument hardware used to handle the connection between the underlying communication protocol;
- (4) Data collection and distribution module: through the network to achieve the same with the student, teacher the same interaction, complete hardware resources and test data of network sharing.

4.3. The Teacher Station Software

Teachers station software mainly includes the following sections:

- (1) Campus network client: laboratory network and campus network interconnection, complete the teaching resources and the related data upload download;
- (2) Laboratory attendance management module: realize the experimental teaching management informatization, accurate record students' attendance information, such as in place, time to leave, and absences information;
- (3) Experimental process regulation module: in the process of experiment, the teacher can be obtained by students of the same real-time operating information, realize the experiment process supervision and management;
- (4) Lab report management module: complete the collection of the electronic experiment report, corrects, check, etc.

4.4. Students Station Software

Students station software mainly includes the following sections:

- (1) Virtual instrument operation modules: mainly provides a user interface and virtual instrument, virtual instrument designed by the loading master station a file, provide students with a virtual instrument operation and running environment;

(2) Experimental report generation module: on the basis of traditional experiment report processing, information processing implementation experiment, experiment report generation and information query functions;

(3) In the experiment, experiment guidance module: built-in relevant theoretical knowledge and experimental frequently encountered problems to solve, is the guarantee the experimental teaching effect is an important tool;

(4) Experimental data record module: used for the experimental operation information in the process of recording and playback experiments and experimental data and other information.

4.5. Distributed Interactive Software

Distributed interactive software mainly displays: the characteristics of distribution and interactivity, heterogeneity, spatial and temporal consistency and openness. Applied in the virtual instrument laboratory system distributed interactive software can greatly promote the function of laboratory network extension, in the same geography position, function and can be distributed computing power, between the same interaction ability. Using distributed interactive technology, make the virtual laboratory constitute an open architecture, each student the same can be arbitrarily and easily to join or leave the system, the dynamic change does not affect the normal operation of the whole system.

5. Conclusion

Build virtual instrument laboratory based on distributed interactive network has a broad prospects for development, in the application of advanced simulation technology in electrical and electronic experimental teaching, a lifelike instrument operation platform for the students to realize the hardware resource sharing, greatly improving the efficiency of the hardware equipment, and laboratory features strong extensibility, virtual instrument build cycle is short, can effectively solve the experiment site in short supply, equipment update such outstanding problems as slow.

Acknowledgement

Fund Project: Anhui University Natural Science Research Project in 2019 [KJ2019A0907]; Major Teaching and Research Project of Anhui Quality Engineering (2017 JYXM 0877); Anhui Wenda Institute of Information Engineering Key Natural Science Foundation Project [XZR2018A02]; 2018 National Undergraduate Innovation and Entrepreneurship Training Program [201812810021].

References

1. Yan Hanmin. Analysis of the construction of laboratory team[J]. Hubei education,2005,(4): 53-54.
2. Zhou Beming, Wan Jufang. The staff team construction is the key of university laboratory construction[J]. The experiment technology and management, 2005,22(8): 6-7.
3. Bian Xiulan. Problems and countermeasures of the laboratory team construction in colleges and universities[J]. Laboratory science, 2006,(5): 138-139.
4. Yan Qihua. The laboratory construction project whole course management[J]. Laboratory research and exp-loration,2004,22(2): 80-82.